

Example sub-graph with security frontend

For "Buffer overflow in OpenSSL 1.0.1f"

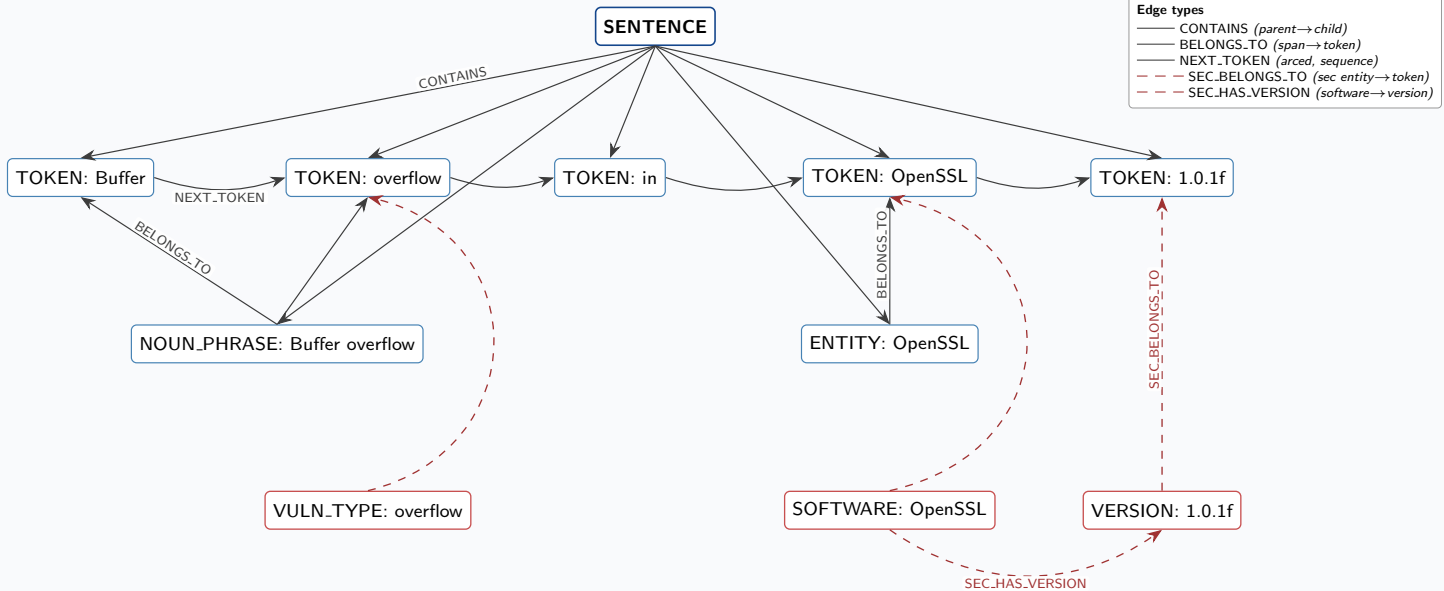


Figure. spaCy produces the blue linguistic layer for "Buffer overflow in OpenSSL 1.0.1f": a SENTENCE root, five TOKEN nodes, the noun phrase "Buffer overflow" and the named entity "OpenSSL", wired together by CONTAINS, BELONGS.TO and NEXT_TOKEN edges. A second pass scans the same text with a small library of regular expressions and dictionaries and adds three red security nodes: the vulnerability-type dictionary matches "overflow" and gives a VULN.TYPE, the software dictionary matches "OpenSSL" and gives a SOFTWARE, and the version regex matches "1.0.1f" and gives a VERSION. Each red node hooks into its source TOKEN through a dashed SEC.BELONGS.TO arrow; the VULN.TYPE and SOFTWARE arrows curve east so they do not cross the linguistic span boxes that sit directly above them. A final dashed arc, SEC.HAS.VERSION, records that this SOFTWARE and this VERSION belong together as one software-version pair.